

Facial Recognition Attendance



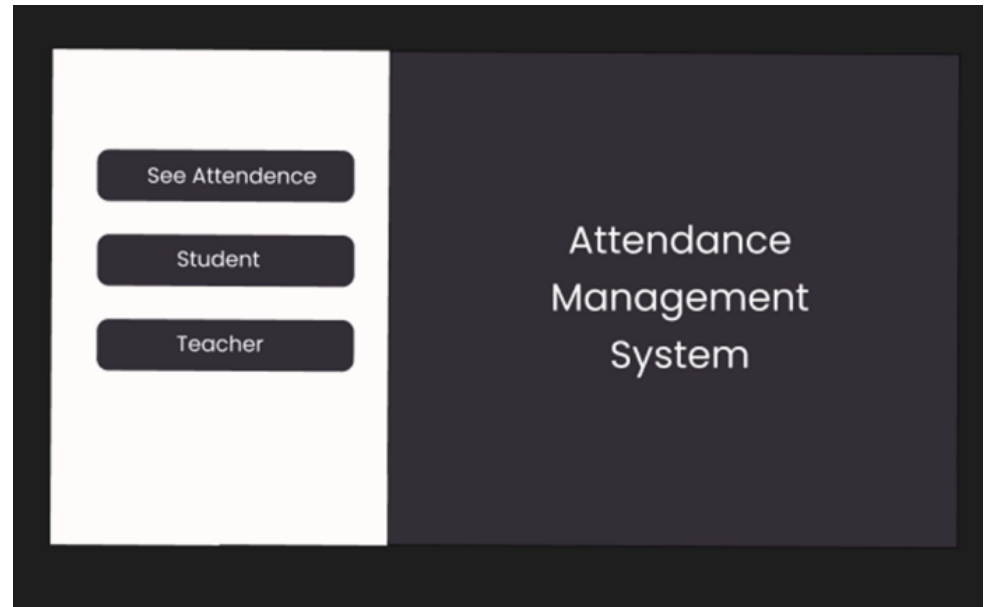
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Face Detection Attendance System

- An automated system using facial recognition to mark attendance.

Introduction

- Traditional attendance systems are time-consuming and prone to proxy attendance.
- Face recognition offers a contactless and efficient solution.

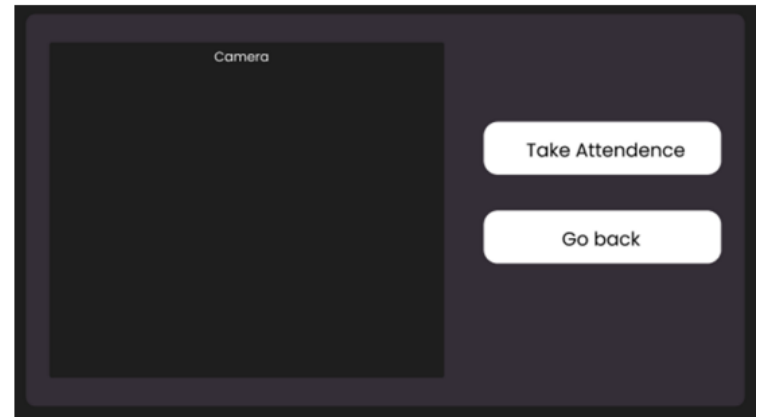


Technology Stack

- • Python
- • OpenCV
- • LBPH (Local Binary Pattern Histogram) algorithm
- • Custom dataset

Face Detection

- Detects faces in images using Haar Cascades or DNNs from OpenCV.
- Extracts grayscale face regions for recognition.



Face Recognition

- • Implemented using LBPH algorithm from OpenCV's cv2.face module.
- • Captures unique local features for each face.
- • Effective in varying lighting conditions.

Attendance Dashboard

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Dataset Preparation

- Custom dataset created with multiple images per person.
- Images converted to grayscale and labeled appropriately.

Model Training

- • LBPH recognizer trained using `cv2.face.LBPHFaceRecognizer_create()`.
- • Input: Labeled grayscale face images.
- • Output: Trained model to recognize known faces.

Attendance Marking

- • Recognized faces are matched with registered users.
- • Attendance marked automatically in a database or file.
- • Timestamp recorded.

Advantages

- • Contactless and automatic attendance.
- • Reduces proxy and manual errors.
- • Scalable for large organizations.

Conclusion

- The LBPH-based face recognition attendance system provides a reliable, efficient, and user-friendly method to automate attendance tracking.